

AI-Assisted Triage System Requirements

Week 9 Final Submission - HCI nurse-facing ED triage dashboard

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System choice: responsive web interface on the existing ED triage workstation. The model provides decision support only; a trained nurse reviews and finalises every ESI decision.

1. System vision

The proposed system is a nurse-facing clinical decision-support interface for the main emergency-department triage desk. It receives approved pre-triage data, validates the inputs, runs the pinned triage model and presents a proposed Emergency Severity Index (ESI) level from 1 to 5. The nurse reviews the recommendation, confirms or overrides it, and remains accountable for the final clinical decision. The system is intended for research and a silent pilot; it shall not diagnose, prescribe treatment or independently prioritise care.

2. Implementation choice

A responsive web interface was selected instead of a mobile application or physical robot for the current pilot. It can run on the existing hospital workstation or an approved tablet, is easier to integrate with the EHR, avoids additional physical hazards near patients and supports a clear manual fallback. The HRI robotic kiosk remains a future comparison concept for the Observation Unit.

3. Intended users and stakeholders

- Primary user: trained ED triage nurse.
- Secondary clinical users: charge nurse and ED physicians reviewing escalation or audit events.
- Technical stakeholders: Clinical IT, cybersecurity, biomedical engineering and data governance.
- Operational stakeholders: Sister Patrice Alleyne, the Integration Review Board, quality and hospital administration.
- Indirect users: patients and caregivers affected by the triage workflow.

4. Workflow

Stage	System action	Human action	Safety rule
Select encounter	Retrieves approved patient and pre-triage data.	Nurse confirms patient and encounter.	No inference before identity and source are visible.
Validate inputs	Checks required fields, range, source and freshness.	Nurse corrects or switches to manual mode.	Inference is blocked when required inputs are missing or invalid.
Generate recommendation	Runs the pinned model and presents proposed ESI 1-5.	Nurse reviews output and patient presentation.	Output is labelled decision support, not a final decision.
Finalise decision	Stores recommendation and nurse action.	Nurse confirms or overrides and documents final ESI.	No automatic finalisation or treatment action.
Audit/review	Logs events and produces aggregate review data.	Governance team reviews performance and override patterns.	Patient-level data remains protected and access controlled.

5. Inputs

Input category	Primary route	Fallback	Validation
Patient/encounter	Read-only EHR pull	Manual encounter selection	Identity, encounter and timestamp visible
Vital signs	EHR/device data already approved for use	Manual nurse entry	Required-field, range, freshness and source checks
Chief complaint	EHR structured field or nurse selection	Manual selection	Plain-language display and source recorded
Approved clinical features	EHR pull	Manual entry only when authorised	No leakage variables or post-outcome data
User identity	Hospital authentication	No anonymous clinical mode	Role and session validity checked

6. Outputs and required human action

Output	Presentation	Required nurse action
Proposed ESI level	Number 1-5 plus urgency text, icon and colour	Compare with clinical presentation; confirm or override
Confidence state	High/moderate/low or manual-review state	Perform fuller review when confidence is low
Missing/invalid data	Persistent warning with affected fields	Correct data or use manual triage
Key contributing observations	Short, non-causal summary of important inputs	Use only as support; do not treat as diagnosis
Audit confirmation	Final ESI, user, time, source and override reason	Verify and save final decision

7. Functional requirements

ID	Testable requirement
FR-01	The system shall require authenticated staff access before identifiable patient data is displayed.
FR-02	The system shall support an approved EHR pull as the primary input route and manual entry as fallback.
FR-03	The system shall display the source and timestamp of each imported clinical input.
FR-04	The system shall validate required fields and block inference when required data is missing or invalid.
FR-05	The system shall display a proposed ESI classification within 1.5 seconds after valid submission.
FR-06	The recommendation shall include the ESI number, urgency text, icon, colour and confidence state.
FR-07	The nurse shall be able to confirm or override the recommendation from the same decision screen.
FR-08	An override shall require a final ESI level and selectable reason, with optional free-text note.
FR-09	The system shall offer manual mode within two user actions when EHR, connectivity or model service fails.
FR-10	The system shall log the model output, final nurse decision, user, timestamp, source and override reason.
FR-11	The system shall never auto-finalise an ESI level, order treatment or delay access to manual triage.
FR-12	The system shall display a persistent statement that nurse judgement is final.

8. Non-functional requirements

ID	Testable requirement
NFR-01	A trained nurse shall identify the proposed ESI and missing-data state within 3 seconds.
NFR-02	The interface shall remain legible under fluorescent lighting, shadows and variable ambient conditions.
NFR-03	Critical meaning shall use text, icon and shape in addition to colour.
NFR-04	The interface shall provide large touch targets and keyboard navigation.
NFR-05	The interface shall use concise, neutral and non-blaming language.
NFR-06	Routine audio alerts shall not be used; approved audio shall be limited to exceptional states.
NFR-07	The system shall recover safely after a brief power or connectivity interruption without auto-submitting a decision.
NFR-08	Patient data shall be encrypted in transit and protected through role-based access.
NFR-09	The system shall avoid exposing patient data in browser logs, screenshots or public telemetry.
NFR-10	The system shall record system faults and maintain an auditable event trail.

9. Integration requirements

ID	Testable requirement
IR-01	The system shall connect to the Mercer EHR through a read-only approved interface such as FHIR R4 or a Mercer-approved API.
IR-02	The system shall use hospital authentication and approved audit-log storage.
IR-03	The workstation shall operate on hospital network controls and approved generator/UPS support.
IR-04	The system clock shall synchronise with the hospital time source.
IR-05	Manual entry shall remain available when the EHR connection is unavailable.
IR-06	The interface shall clearly indicate online, offline and degraded modes.
IR-07	No robot, sensor or device stream shall be added without validation, biomedical approval and a manual fallback.

10. Safety considerations

Setting	Concern	Context	Mitigation	Residual risk
HCI	Alarm fatigue	Busy ED with existing alarms and interruptions	Visual-first design, duplicate suppression and monitored thresholds	Local alert thresholds may still be miscalibrated
HCI	Legibility under stress	Variable light, clutter and fatigue	High contrast, large type and one primary action	User testing is still needed
HCI	Accessibility	Colour-only meaning and small controls can exclude users	Text + icon + colour, keyboard access and scalable text	Unidentified needs may emerge in testing
HRI	Voice failure	Noise, masks, accents and multilingual speech	Voice optional; visible confirmation; touch/manual alternative	Voice may remain unusable for some users
HRI	Sensor failure	Dropout or partial data may mislead classification	Block inference, alert nurse and switch to manual mode	Manual response may be delayed
HRI	Physical proximity	Robot may obstruct, startle or create a collision risk	Stationary pilot, clearance zone and emergency stop	On-site physical risk assessment remains essential

11. Failure modes and graceful degradation

Failure event	System response	Human response	Audit evidence
Power loss	Screen goes dark; no decision is submitted	Nurse follows manual triage protocol	Power event logged when service returns
Connectivity loss	Offline banner; EHR pull disabled	Nurse uses manual entry	Connection loss and source change logged
Missing/invalid data	Inference blocked; affected fields highlighted	Nurse corrects values or assesses manually	Validation failure logged
Low confidence	Manual-review state shown	Nurse performs full clinical review	Confidence state and final decision logged
User disagreement	Override remains visible	Nurse selects final ESI and reason	Both recommendation and override logged
Model service failure	No recommendation shown	Nurse continues manual triage	Service failure logged

12. Privacy, governance and audit

- Use only the minimum necessary patient information for the active encounter.
- Apply encryption in transit, secure sessions, role-based access and automatic logout.
- Do not store patient data in browser console logs, screenshots or public analytics.
- Record who created or changed each encounter, source of each input, model output, final ESI and override reason.
- Review fairness, calibration, missing-data behaviour, critical-class performance and drift before operational deployment.
- Treat the system as research decision support, not as a certified clinical device.

13. Known limitations and next steps

- The current model has not been prospectively validated at Mercer or in a Caribbean ED.
- Performance for rare critical ESI cases remains uncertain and must be reviewed separately.

- Patient-level explanations require further design and clinical validation.
- The mock-up has not yet been tested with real nurses, accessibility users or near-miss scenarios.
- Week 10 should test failure cases, override behaviour, alarm fatigue and manual fallback.

Source basis

This document follows the Week 9 CariSurg tutorial structure for inputs, outputs, required human action, functional/non-functional/integration requirements, co-design canvases and the Concern-Context-Mitigation-Residual Risk safety format.